Supplementary Material: Anchor the Head: Alignment Drift in Open-Vocabulary Segmentation

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This supplement records reproducibility details for the experiments reported in the main paper. The main paper is self-contained. The material here clarifies how the released JSON files, split files, and scripts support staged reproduction. The public repository contains code, splits and result records.

Evaluation protocol per table

Different comparisons use different image sets and resolutions. Table S0 lists the protocol behind every table in the paper and in this supplement, so same-method numbers can be compared across tables.

Table S0: Evaluation protocol behind each table. *Masked-crop*: frozen SAM+CLIP, painted and scored at 256×256, on the SAM-cacheable subset of the 2000 ADE20K-150 val images (original cache with the image count not recorded at build time, independent rebuild of 1623 images). *Dense-CLIP*: no proposer, all 2000 val images, scored at the backbone's patch resolution.

Table	Stack and data	Images	Splits
1	ADE20K masked-crop	cacheable subset	seed 0
2, 4, 5	ADE20K dense-CLIP	2000	5
3	VOC2012 dense-CLIP	VOC val	5
S1, S5	ADE20K masked-crop	347	5
S3	ADE20K masked-crop	347	seed 0
S2, S4	ADE20K masked-crop	cacheable subset	as Table 1
S6, S7, S8	ADE20K dense-CLIP	2000	5
S10	Dense-CLIP timing	GPU: 2000; CPU: 100	1 (training)
S9	COCO fit, ADE dense eval	600 fit, 2000 eval	whole ADE vocabulary

A Reproducibility

Model inference and head training use a single GPU and fine-tune neither frozen model. We use a frozen SAM mask proposer and a frozen CLIP vision–language model and evaluate on ADE20K-150 validation. The masked-crop pipeline can only score images for which SAM emits at least two proposals, so the headline accuracy tables evaluate the SAM-cacheable subset of the 2000-image validation split rather than all 2000 images (the independent rebuild cached 1623 images; see `config.n_val` in `results/rebuild_mc/metrics_calib.json`). The original cache behind Table 1 and Tables S2 and S4 has no recorded image count. Calibration and encoder comparisons use 347 validation images over five splits and 300 training images; the edge ablation uses seed 0 only. Dense-CLIP scores all 2000 validation images at patch resolution. Each ADE split holds out 30 of 150 classes. Masked-crop splits use `default_rng(seed)` for seeds 0–4; dense splits use five successive permutations from one `RandomState(0)`

stream. Means and standard deviations are over splits where indicated; bootstrap intervals resample images at seed 0. All methods start from the *identical* frozen-CLIP per-mask logits, so each comparison isolates the reasoning or calibration layer from the score. Post-hoc calibrators (global temperature, per-class temperature, vector scaling) are fit only on seen-class training masks, exactly the supervision a deployed open-vocabulary system would have. The proposal graph is built from mask IoU, boundary overlap, and positive CLIP-region cosine similarity, and diffusion mixes the frozen CLIP logits back at every step, a class-symmetric and parameter-free operation. Reproduction is staged by cost in `REPRODUCE.md`. JSON-only generators rebuild Tables 1, 2, 4, 5, S2, S4 and S6–S10, with manuscript-specific captions and layout. The S1/S3 writers require a feature cache, and Table 3 and S5 are transcribed from records. Masked-crop calibration runs with CPU `numpy`; dense-CLIP and VOC evaluation require CUDA and separate dense caches (`MODELS.md`). Five masked-crop ADE files (`ade20k_seed*`), five dense ADE files (`ade20k_dense_seed*`), and five VOC files (`voc_seed*`) are provided under `splits/` with checksums.

Additional results

The result records and generation scripts are described in `scripts/EXTENDED.md` in the public repository.

B Full calibration study

Table S1: Post-hoc calibrators on identical frozen-CLIP per-mask logits. ADE20K-150 val, N=347 images, five seen/novel splits.

Method	mIoU	novel-IoU	ECE↓	novel ECE↓	seen ECE↓
Uncalibrated baseline	$8.9_{\pm 0.0}$	$8.4_{\pm 0.9}$	$14.6_{\pm 0.0}$	$12.8_{\pm 3.0}$	$15.3_{\pm 0.8}$
Temperature scaling (global T)	$8.8_{\pm 0.0}$	$8.5_{\pm 0.8}$	$3.5_{\pm 0.6}$	$6.6_{\pm 3.1}$	$3.1_{\pm 0.8}$
Per-class temperature	$3.6_{\pm 0.6}$	$0.5_{\pm 0.6}$	$12.6_{\pm 2.8}$	$27.9_{\pm 3.5}$	$8.9_{\pm 2.5}$
Vector/Platt scaling	$3.7_{\pm 0.6}$	$0.0_{\pm 0.0}$	$16.2_{\pm 0.7}$	$34.0_{\pm 4.4}$	$12.0_{\pm 1.6}$
Graph diffusion	$9.1_{\pm 0.0}$	$8.5_{\pm 1.0}$	$5.9_{\pm 0.0}$	$5.2_{\pm 1.7}$	$6.6_{\pm 0.9}$

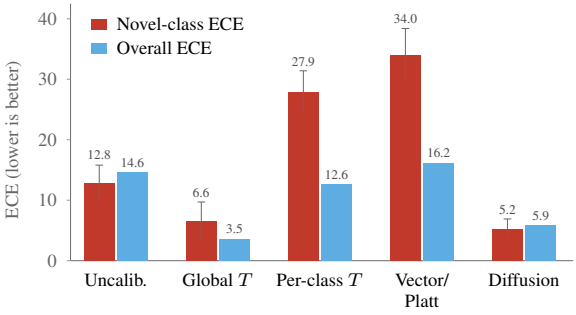


Figure S1: Novel-class (red) and overall (blue) ECE on identical frozen-CLIP per-mask logits, ADE20K-150 val. Values from Table S1. Lower is better.

C Diffusion sweep and edge ablation

Table S2: Diffusion sweep over weight α and steps k. ADE20K-150 val, original masked-crop cache. The α=0 row is the independent Proposal+CLIP baseline.

Config	mIoU	hIoU	novel-IoU	ECE↓
baseline (α=0)	8.9	9.1	9.4	14.6
α=0.2, k=1	8.8	9.0	9.4	9.3
α=0.2, k=2	8.7	8.9	9.2	9.8
α=0.2, k=3	8.7	8.9	9.2	9.7
α=0.4, k=1	8.6	9.1	10.1	4.5
α=0.4, k=2	**9.0**	**9.4**	10.0	5.9
α=0.4, k=3	8.9	9.3	10.0	5.4
α=0.6, k=1	8.1	8.6	9.8	2.4
α=0.6, k=2	8.9	9.2	9.8	3.5
α=0.6, k=3	8.5	9.1	**10.5**	**2.2**
α=0.8, k=1	7.0	7.5	8.7	4.6
α=0.8, k=2	**9.0**	9.3	10.0	4.5
α=0.8, k=3	7.7	8.1	9.0	3.0

Table S3: Edge-source ablation for graph diffusion (α=0.4, k=2), ADE20K-150 val, 347 images, seed 0. Each row restricts the diffusion edge weight to a single evidence source.

Edge source	mIoU	novel-IoU	ECE↓	novel ECE↓
all edges	9.1	10.2	5.9	5.5
IoU only	8.7	9.6	6.4	5.9
boundary only	9.0	10.5	5.9	5.3
CLIP-cosine only	9.1	10.2	5.8	5.4

D Full decoder ablation

Table S4: Decoder ablation on ADE20K-150 val, original masked-crop cache. Rows remove the graph, calibration, or prototype-refinement component of the trained decoder.

Variant	mIoU	seen-IoU	novel-IoU	hIoU	ECE↓
Proposal+CLIP baseline	8.9	8.7	9.4	9.1	14.6
full (graph+residual+calib+proto)	7.1	8.6	1.1	2.0	23.8
− graph (per-node MLP)	8.1	9.8	1.5	2.7	24.6
− calibration head	7.1	8.6	1.3	2.2	23.6
− prototype refinement	4.3	5.5	0.0	0.0	27.8

E Stronger-encoder probe

Table S5: Stronger-encoder probe on ADE20K-150 val masked crops: a context-padded region encoder (pad 0.25) against the standard crop, with calibration baselines.

Encoder	Method	mIoU	novel-IoU	ECE↓	novel ECE↓
bbox crop (weak)	baseline	8.9	8.4	14.6	12.8
	global T / diffusion	8.8 / 9.1	8.5 / 8.5	3.5 / 5.9	6.6 / **5.2**
context crop (strong)	baseline	9.2	9.3	16.0	13.8
	global T / diffusion	9.3 / 9.3	9.4 / 9.3	3.8 / 6.9	8.2 / **7.7**

F PASCAL VOC regime

Table 3 uses frozen CLIP ViT-B/16 with Correlative Self-Attention on VOC2012 validation, at patch resolution (75.3 mIoU; the recorded full-resolution base is 81.8). Six of the 20 object classes are novel in each of five sequential permutations from `RandomState(0)`. Background and void labels are excluded. Corrections are fitted on seen-class patches from the training split. The released `results/metrics_voc.json` gives split variation: novel-class IoU is 75.2 ± 8.0 for the frozen base and global temperature, 0.0 ± 0.0 for per-class temperature, 77.1 ± 8.1 for diffusion, and 16.1 ± 5.5 for the decoder. Global-temperature ECE is 4.3 ± 0.4.

G Class-symmetric vs trained baselines with AURC

Table S6: Class-symmetric smoothing and calibration baselines on the ADE20K-150 dense-CLIP base, five seen/novel splits, with AURC alongside IoU and ECE.

Method	mIoU	hIoU	ECE↓	novel-IoU	AURC↓
Graph diffusion (anchored)	15.0	15.2	27.8	15.7	49.2
Graph Laplacian (generic, no anchor)	15.5	15.8	28.5	16.3	48.6
Trained MLP (feature input)	5.5	0.2	2.6	0.1	38.2
Trained decoder (logit-space TR)	10.5	7.6	11.0	5.8	34.2
Text-conditioned per-class calibrator	8.0	3.2	19.0	1.9	–

H Frontier grid

Table S7: Frontier grid on the ADE20K-150 dense-CLIP base, five splits. g mixes the frozen and corrected *probabilities* (B3). η scales the learned *logit residual* (B4). 0 is the frozen baseline, 1 the unanchored head. C is novel-IoU divided by frozen-baseline novel-IoU.

Setting	mIoU	seen-IoU	novel-IoU	hIoU	ECE↓	C
frozen baseline	$14.5_{\pm 0.0}$	$14.3_{\pm 0.3}$	$15.1_{\pm 1.0}$	$14.7_{\pm 0.4}$	$27.2_{\pm 0.0}$	1.00
B3 output gate g=0.1	$15.9_{\pm 0.3}$	$16.2_{\pm 0.3}$	$14.7_{\pm 1.4}$	$15.4_{\pm 0.8}$	$36.5_{\pm 1.6}$	0.97
B3 output gate g=0.2	$15.1_{\pm 0.4}$	$15.7_{\pm 0.4}$	$13.0_{\pm 2.1}$	$14.1_{\pm 1.3}$	$32.8_{\pm 2.1}$	0.86
B3 output gate g=0.4	$12.4_{\pm 0.5}$	$13.6_{\pm 0.4}$	$8.0_{\pm 2.5}$	$9.8_{\pm 2.0}$	$23.3_{\pm 2.3}$	0.53
B3 output gate g=0.6	$9.6_{\pm 0.5}$	$11.0_{\pm 0.4}$	$3.8_{\pm 2.1}$	$5.3_{\pm 2.5}$	$14.0_{\pm 2.3}$	0.25
B3 output gate g=0.8	$7.2_{\pm 0.3}$	$8.9_{\pm 0.3}$	$0.6_{\pm 0.6}$	$1.1_{\pm 1.0}$	$5.3_{\pm 2.0}$	0.04
B3 output gate g=1.0	$6.5_{\pm 0.3}$	$8.1_{\pm 0.3}$	$0.1_{\pm 0.1}$	$0.1_{\pm 0.3}$	$3.9_{\pm 2.5}$	0.01
B4 residual scale η=0.0	$14.5_{\pm 0.0}$	$14.3_{\pm 0.3}$	$15.1_{\pm 1.0}$	$14.7_{\pm 0.4}$	$27.2_{\pm 0.0}$	1.00
B4 residual scale η=0.25	$14.2_{\pm 0.4}$	$15.5_{\pm 0.5}$	$9.3_{\pm 2.3}$	$11.4_{\pm 1.8}$	$29.7_{\pm 2.7}$	0.61
B4 residual scale η=0.5	$11.2_{\pm 0.3}$	$13.1_{\pm 0.5}$	$3.5_{\pm 1.9}$	$5.3_{\pm 2.5}$	$19.2_{\pm 1.9}$	0.23
B4 residual scale η=0.75	$8.1_{\pm 0.3}$	$10.0_{\pm 0.4}$	$0.5_{\pm 0.5}$	$0.8_{\pm 0.8}$	$7.3_{\pm 2.1}$	0.03
B4 residual scale η=1.0	$6.5_{\pm 0.3}$	$8.1_{\pm 0.3}$	$0.1_{\pm 0.1}$	$0.1_{\pm 0.3}$	$3.9_{\pm 2.5}$	0.01

I Confound ablation

Table S8: One-factor-at-a-time variations of the seen-only head on the ADE20K-150 dense-CLIP base, five seen/novel splits. Each row changes one factor. The C4 rows sweep the trust-region bound τ. C is novel-IoU divided by frozen-baseline novel-IoU; drift is mean cosine distance of probability vectors, multiplied by 100.

Condition	mIoU	seen-IoU	novel-IoU	hIoU	ECE↓	drift	C
Uncalibrated (frozen baseline)	$14.5_{\pm 0.0}$	$14.3_{\pm 0.3}$	$15.1_{\pm 1.0}$	$14.7_{\pm 0.4}$	$27.2_{\pm 0.0}$	0.0	1.00
Seen-only head (CE)	$6.5_{\pm 0.4}$	$8.2_{\pm 0.5}$	$0.1_{\pm 0.2}$	$0.2_{\pm 0.4}$	$3.9_{\pm 2.7}$	56.9	0.01
C1 class-balanced CE	$9.6_{\pm 0.5}$	$12.0_{\pm 0.6}$	$0.0_{\pm 0.0}$	$0.0_{\pm 0.1}$	$16.5_{\pm 3.0}$	36.3	0.00
C2 focal loss	$7.0_{\pm 0.5}$	$8.7_{\pm 0.5}$	$0.1_{\pm 0.1}$	$0.1_{\pm 0.3}$	$4.5_{\pm 1.5}$	54.7	0.01
C2 cosine-margin loss	$4.9_{\pm 0.3}$	$6.0_{\pm 0.4}$	$0.1_{\pm 0.1}$	$0.1_{\pm 0.3}$	$20.3_{\pm 3.7}$	63.0	0.01
C3 learned logit scale	$7.8_{\pm 0.4}$	$9.7_{\pm 0.5}$	$0.3_{\pm 0.4}$	$0.6_{\pm 0.7}$	$5.2_{\pm 2.4}$	55.5	0.02
C4 trust region τ=0.5	$15.2_{\pm 0.4}$	$15.4_{\pm 0.5}$	$14.6_{\pm 0.9}$	$14.9_{\pm 0.5}$	$31.2_{\pm 1.9}$	5.6	0.96
C4 trust region τ=1.0	$14.3_{\pm 0.3}$	$14.8_{\pm 0.3}$	$12.1_{\pm 2.1}$	$13.2_{\pm 1.2}$	$28.2_{\pm 2.0}$	19.4	0.80
C4 trust region τ=2.0	$10.6_{\pm 0.4}$	$11.7_{\pm 0.4}$	$5.9_{\pm 2.1}$	$7.6_{\pm 1.9}$	$10.9_{\pm 1.6}$	44.6	0.39
C4 trust region τ=4.0	$6.8_{\pm 0.4}$	$8.3_{\pm 0.5}$	$0.5_{\pm 0.6}$	$0.9_{\pm 1.0}$	$3.3_{\pm 2.4}$	55.8	0.03
C5 supervision breadth: 30 seen classes	$3.7_{\pm 2.6}$	$3.8_{\pm 2.2}$	$3.3_{\pm 4.3}$	$3.1_{\pm 3.5}$	$22.2_{\pm 12.5}$	63.7	0.22
C5 supervision breadth: 60 seen classes	$2.7_{\pm 0.2}$	$3.3_{\pm 0.3}$	$0.2_{\pm 0.3}$	$0.4_{\pm 0.5}$	$19.0_{\pm 6.5}$	67.3	0.01
C5 supervision breadth: 90 seen classes	$4.3_{\pm 0.5}$	$5.4_{\pm 0.6}$	$0.1_{\pm 0.1}$	$0.1_{\pm 0.2}$	$12.8_{\pm 3.8}$	62.7	0.00
C5 supervision breadth: 120 seen classes	$6.5_{\pm 0.4}$	$8.2_{\pm 0.5}$	$0.1_{\pm 0.2}$	$0.2_{\pm 0.4}$	$3.9_{\pm 2.7}$	56.9	0.01
C6 spatially pure patches only	$4.8_{\pm 0.1}$	$5.9_{\pm 0.2}$	$0.2_{\pm 0.2}$	$0.4_{\pm 0.4}$	$4.0_{\pm 2.2}$	57.5	0.01

J COCO-Stuff transfer

Table S9: Corrections fit on COCO-Stuff-171 patches from 600 COCO val2017 images (same frozen backbone and fitting budget as in-domain), applied to all 2000 ADE20K-150 val images.

Correction (fit on COCO-Stuff)	mIoU	acc	ECE↓	drift
None (frozen baseline)	14.5	35.4	27.2	0.0
Global temperature	14.5	35.4	8.4	13.7
Text-conditioned calibrator	14.3	43.1	6.5	33.8
Class-agnostic emb. head (λ=0)	14.9	56.4	4.6	46.1
Class-agnostic emb. head (λ=1)	15.6	55.8	8.9	41.6
Diffusion (fit-free)	15.0	35.9	27.8	0.2
Per-class T / logit-space head: *not applicable* (parameters indexed by COCO classes)				

K Cost

Table S10: Compute cost of each correction (dense-CLIP regime, single H100 MIG 2g.20gb slice). Graph: 576 nodes and 4608 kNN entries per image on average (k_{nn}=8). 341 004 training patches; one training split. GPU diffusion scores 2000 images, CPU diffusion 100. Timing includes graph construction and device transfers; CPU-row GPU memory is resident allocation.

Component	wall clock	per image	peak GPU mem
Graph diffusion (GPU, full val)	1.8 s	0.9 ms	259 MB
Graph diffusion (CPU, 100 imgs)	1.4 s	14 ms	242 MB
Global temperature fit	1.5 s	–	1475 MB
Decoder training (12 ep, 1 split)	1.6 s	–	346 MB
Decoder parameters: 169 878			

L Full specification of the audited corrections

Dense residual heads. The dense-CLIP heads use a two-layer MLP $\mathbb{R}^{512} \to 256 \to \mathbb{R}^{150}$ with ReLU, applied to patch features. Frozen logits have scale 40. Tables 2 and S6 use a 0.5 output multiplier for the feature-input MLP; the Table 4 alignment family adds the unscaled logit residual. B2 instead predicts a 512-dimensional embedding residual. Training uses Adam, learning rate 10^{-3}, batches of 8192 patches and 12 epochs, with cross-entropy on seen-class patches from the 600-image train cache. No novel labels are used for stopping or selection. Frozen-logit distillation adds $\lambda T_{\mathrm{KD}}^2 \mathrm{KL}(\sigma(\ell^0/T_{\mathrm{KD}}) \| \sigma(\ell'/T_{\mathrm{KD}}))$ with T_{KD}=2, sampled from all training patches without their labels. Here λ weights distillation; the separate trust-region bound τ clips each logit residual to $[-\tau, \tau]$.

Masked-crop graph decoder. The decoder in Tables 1 and S4 uses the graph, embedding residual, calibration and prototype components of main-paper §4, with 768-dimensional region features, logit scale 100, and a $0.2\tanh$ embedding residual. It trains for 12 epochs with Adam (learning rate 10^{-3}, weight decay 10^{-4}). It is distinct from the dense MLP.

Unseen-class parameters in per-class calibrators. Only seen-class parameters are updated. In the masked-crop calibration runs (S1, S5 and Fig. S1), novel classes inherit the fitted global temperature: $T_c = T_{\mathrm{global}}$, or $a_c = 1/T_{\mathrm{global}}, b_c = 0$ for vector scaling. Dense-CLIP and VOC per-class temperature runs instead leave novel classes at $T_c = 1$. These conventions both avoid fitting novel-class parameters but use different reference scales.

Confidence and ECE. Masked-crop ECE is computed once per labeled proposal against its majority ground-truth label, over all proposals whose label is not ignored, with 15 equal-width confidence bins. It is computed before painting and does not depend on which overlapping proposal wins a pixel. Pixel IoU uses the highest-confidence covering proposal. Dense ECE uses labeled patches directly. Adaptive-bin and class-balanced dense ECE variants are recorded separately in the artifact.

Diffusion hyperparameters. α=0.4 and k=2 were fixed before novel-class evaluation. Table S2 reports the full weight/step sweep, without selecting on novel labels. In contrast, η denotes the residual interpolation scale in B4; it is unrelated to the diffusion weight.

Dense graph and learned affinity. Per image, the control uses a cosine kNN graph (k_{nn}=8) on normalized CLIP patch features, excludes self-edges, clips negative similarities, symmetrizes the weights, and row-normalizes them (denominators clamped at 10^{-6}). With the resulting matrix A, diffusion iterates $P \leftarrow (1-\alpha)P_0 + \alpha AP$ for k steps, on probability vectors. The generic Laplacian drops the anchor term. B6 learns graph affinities within this anchored update; it is not an unanchored control. The cost record averages 576 nodes and 4608 directed neighbor entries per image. LPOSS uses a separate DINO graph with k_{nn}=400.

Drift accounting. Drift is 100 times the mean cosine distance between corrected and frozen class-probability vectors over labeled patches. The 38 class-specific sweep settings include the frozen B4 endpoint and repeated unanchored endpoints: 32 distinct correction configurations remain after excluding the frozen endpoint and retaining one unanchored-head setting. The reported pointwise and family-bootstrap statistics use all 38 measured settings; excluding the frozen endpoint and repeated head configurations gives approximately $r = -0.88$. The fit in Fig. 4 uses those 38 settings, excluding the class-symmetric controls.

Independent rebuild. Reproduction deviations are computed by `scripts/repro_deviation.py` and stored in `results/extended/repro_deviation.json`. Compared dense rows cover the uncalibrated base, global and per-class temperatures, diffusion and Laplacian, agreeing within 0.05; temperatures are fitted, not parameter-free. Trained-decoder metrics differ by at most 0.85 (VOC novel ECE 16.7 versus 17.5), with ADE maximum 0.40. Trained-head AURCs are undefined in the rebuild and excluded. There is no independent rebuild record for the alignment/frontier, diagnostic, transfer or LPOSS experiments (Tables 4, 5, S7–S9). The masked-crop rebuild reproduces the qualitative pattern at 10.2 baseline mIoU on 1623 cacheable images. A repeated run on that rebuilt cache agrees within 1.2 for decoder-ablation metrics and within 0.02 for five-split calibration. Absolute results from independently generated SAM caches are not directly comparable.

M Claims and scope

Table S11: Claim-to-evidence and scope map. Protocol in §A. "Dense regime": training-free dense-CLIP (14.5 mIoU at patch resolution). "Masked-crop regime": frozen SAM+CLIP, two caches (8.9 and 10.2 mIoU). Each entry names its cache.

Claim	**Evidence**	**Scope / not claimed**
Seen-only, class-specific, unanchored corrections collapse novel classes.	Novel-IoU $15.1 \rightarrow 0.1$ (dense, Tab. 4). $9.4 \rightarrow 1.1$ on the original masked-crop cache (Tab. 1, Supp. Tab. S4) and $9.5 \rightarrow 2.5$ on its independent rebuild (`results/extended/mc_metrics_protomask.json`). Per-class T and vector scaling alike (Supp. Tab. S1).	Five bases: masked-crop, dense-CLIP, LPOSS, DINO-proxy and VOC. ADE mIoU 9–15.5, VOC 75–82. Masked-crop accuracy is seed 0; dense and calibration comparisons use five splits. CorrCLIP/CASS are untested.
Alignment drift tracks novel-IoU reduction within the class-specific family; class-symmetric controls are exceptions.	Pointwise $r = -0.90$ (cluster-bootstrap CI $[-0.96, -0.74]$). Family-level $r = -0.65$ $[-0.95, -0.23]$ over 13 families (Fig. 4). Global T (drift 13.4) and LPOSS (drift 50.6) retain novel-class IoU. Learned affinity in anchored diffusion (B6) retains it at $C = 1.04$.	Correlational, with hyperparameter curves treated as correlated evidence. An intervention axis, not a causal law or theorem.
Anchoring recovers novel-class IoU: distillation to frozen logits restores novel classes and improves mIoU.	$\lambda = 30$: novel-IoU 15.3 ($C = 1.01$), mIoU $14.5 \rightarrow 15.8$. Monotone in λ, g, η, τ (Tables 4, S7, S8, Fig. 3).	Dense regime, five splits. Anchored heads underpredict novel *pixels* (7–11% vs. 15.6%): IoU is preserved via precision, not coverage.
The collapse is not explained by class imbalance, objective, logit scale, supervision breadth, or label purity.	Each confound varied in isolation. Novel-IoU stays ≤ 0.3 in every setting but the narrowest supervision (30 seen classes: 3.3 ± 4.3, $C = 0.22$, still a large reduction against 15.1). Only the trust region ($\tau = 0.5$: 14.6) rescues it (Table S8).	More seen classes reduce novel-class IoU further. Nothing is claimed for confounds we did not vary.
Global T is the best in-domain calibrator among corrections retaining novel-class IoU. Diffusion/LPOSS are class-symmetric *controls*, not contributions.	ECE $27.2 \rightarrow 8.7$ (dense) with novel-IoU unchanged. Collapsed heads post the *lowest* ECE/AURC (Supp. Table S6). Laplacian $\approx$ anchored diffusion. LPOSS compared under a matched protocol.	ECE/AURC do not flag the collapse – only novel-IoU does. No claim of beating LPOSS or modern OVSS.